

SpeechMate: An AI-Powered Communication and Speech Training Assistant

by

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Interim report submitted in partial fulfilment of
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CERTIFICATION OF APPROVAL

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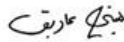
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An interim report submitted to the
Computer Science Programme
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in partial fulfilment of the requirement for the
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Approved by,



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September 2026

CERTIFICATION OF ORIGINALITY

This is to certify that I am responsible for the work submitted in this project, that the original work is my own except as specified in the references and acknowledgements, and that the original work contained herein has not been undertaken or done by unspecified sources or persons.

A handwritten signature in black ink, appearing to read 'Por Jie Hao', is positioned above a horizontal line.

POR JIE HAO

ABSTRACT

This report presents the fundamental progress of SpeechMate, an AI-powered communication and speech training platform designed to address the persistent gap between Malaysian graduates' academic qualifications and their communication competence in professional settings. A national employer survey reported that 64% of hiring managers cited poor English proficiency, and 53% cited a lack of general communication and presentation skills, as the main reasons Malaysian graduates fail interviews and remain unemployed. Existing commercial speech training tools are still limited: they largely limit training to isolated vocabulary drills, require users to stare at static scripts on a screen while speaking (a pattern this project calls “Gaze Tunneling”), and interrupt speakers mid-performance with real-time alerts that increase cognitive load rather than reduce it.

To address this gap, SpeechMate integrates automatic speech recognition (Faster-Whisper), computer vision-based performance behavior analysis (MediaPipe and OpenCV), and a large language model conversation trainer (Gemini Flash) into a single, locally deployed web application. The project is guided by four research objectives: an AI-powered pre-performance script assistant, a real-time dialogue system that assesses suprasegmental as well as segmental speech features, a multimodal data fusion pipeline that links vocal stuttering insights and events on a unified timeline, and a delayed post-session analytics dashboard aligned with Cognitive Load Theory.

This interim report documents progress through the system architecture design phase and the start of AI model training. It presents the finalized background, problem statement, objectives, and scope of the study; a literature review consisting of four theoretical foundations, four closely related technical works, and a summary table of the reviewed works; the Agile-based research methodology, system architecture, use case diagrams, technology stack, project schedule, expected system deliverables, and process flow charts; and the planned data collection, preprocessing, model training configuration, evaluation criteria, and interface design (wireframes and storyboards) for the AI components, derived from the reviewed literature.

The completed SpeechMate system is expected to produce a working web-based prototype that allows users to complete a full training cycle – scenario selection, AI-guided conversation training, and delayed multimodal performance feedback – without any real-time screen interruptions. Expected outcomes include a quantitative reduction in “Gaze Tunneling” behavior compared to a script reading baseline, objective fluency and pronunciation scores benchmarked against the technical assessment framework in Chapter 4, a functional six-module dashboard that can be used by students, educators, and administrators, and evidence, gathered through user acceptance testing, of increased speaking confidence after repeated training sessions. The report concludes with a summary of progress to date and the work planned for the remaining development, integration, and validation phases..

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LIST OF ABBREVIATIONS

AI	Artificial Intelligence
API	Application Programming Interface
ASR	Automatic Speech Recognition
BiLSTM	Bidirectional Long Short-Term Memory
CALL	Computer-Assisted Language Learning
CNN	Convolutional Neural Network
CV	Computer Vision
FYP	Final Year Project
GOP	Goodness of Pronunciation
JWT	JSON Web Token
LLM	Large Language Model
MALL	Mobile-Assisted Language Learning
PDF	Portable Document Format
UAT	User Acceptance Testing
UCLASS	University College London Archive of Stammered Speech
UI	User Interface
UTP	Universiti Teknologi PETRONAS
WER	Word Error Rate
WPM	Words Per Minute

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CHAPTER 1

INTRODUCTION

1.1 Background of Study

Communication skills have become a determining factor in graduate employability in Malaysia. According to the JobStreet.com Malaysia Graduate Employability Survey and Job Outlook Report (2025), 64% of corporate managers and recruitment staff cited poor English proficiency as the main reason fresh graduates failed interviews and remained unemployed. A separate industry assessment by JobStreet Malaysia and the Malaysian Employers' Federation (2025) further reported that 53% of employers cited a lack of communication and presentation skills among local applicants, noting that graduates often struggle to verbally convey their qualifications despite having strong academic records.

Existing computer-assisted language learning (CALL) and mobile-assisted language learning (MALL) applications do not close this gap. Previous evaluations of standard commercial tools have found that, while helpful for vocabulary acquisition, such tools largely limit learners to producing isolated words and memorized phrases rather than functional spontaneous conversation (Loewen, Isbell & Sporn, 2020). Furthermore, tools built specifically for presentation training typically require users to read from an on-screen script while speaking, which reinforces poor eye contact habits and audience disengagement (Kang, Yang, Song, Yang & Wang, 2024). Many active feedback systems also interrupt speakers mid-performance with visual or auditory alerts, which impairs working memory and increases speaking anxiety rather than reducing it, an effect documented in recent research on stress testing of spoken language models (Yosha, Maimon & Adi, 2026).

SpeechMate was proposed as a response to this specific deficiency. It combines automatic speech recognition, computer vision and conversational artificial intelligence into a local web-based platform that allows students, job seekers and individuals with mild speech impairments to practice verbal and non-verbal communication skills in a private, judgment-free environment, and receive structured feedback only after the training session is over.

1.2 Problem Statement

Four specific, literature-supported problems motivate this project.

First, multilingual learners experience significant mental blocks when translating thoughts into formal English under pressure, which triggers vocal anxiety and stuttering, largely because they lack tools to facilitate and rehearse their messages before delivering (Kho & Ting, 2023). Second, standard commercial MALL applications help with vocabulary but limit learners to producing isolated words and memorized phrases rather than fluent spontaneous conversation (Loewen, Isbell & Sporn, 2020). Third, existing training tools require users to look down or stare directly at a static digital script while speaking, reinforcing a habit this project calls “Gaze Tunneling,” which prevents users from practicing natural audience eye contact (Kang et al., 2024). Fourth, most existing training apps use flashing lights, alerts, or instant timers while users are actively speaking; such active distractions damage working memory, disrupt the flow of the presentation, and exacerbate mid-speech anxiety (Yosha, Maimon & Adi, 2026).

Taken together, these four problems indicate that current tools address either the verbal or nonverbal dimensions of communication separately, and even when feedback is offered, its timing and delivery can be counterproductive for anxious speakers. SpeechMate is designed to address all four problems simultaneously.

1.3 Objectives

The project is guided by the following four research objectives:

1. To develop an AI-powered Pre-Presentation Script Assistant that refines rough text input into conversationally fluent presentation scripts, reducing the initial cognitive load of the user.
2. To build a real-time dialogue system that assesses speech proficiency based on suprasegmental features (e.g., rhythm, pitch, intonation) in addition to traditional segmental phonetic accuracy.

3. To implement a multimodal data fusion pipeline that uses webcam gaze tracking to mathematically map vocal stuttering events to visual slide gaze coordinates (Gaze Tunneling) on a consolidated session timeline.
4. To provide a delayed post-session analytics dashboard that ensures silent and alert-free presentation simulations, generating automatic recovery metrics only after speech is complete.

1.4 Scope of Study

This project focuses on the development of AI-powered communication and speech training web applications. Its scope includes automatic speech recognition and analysis, computer vision-based performance-behavior analysis, conversational AI trainers and progress tracking dashboards, and is limited to text-based conversational practice and browser-based delivery on desktop and laptop devices.

1.4.1 Target Audience

- University students preparing for presentations, viva sessions and project demonstrations.
- Job seekers and young professionals preparing for interviews, workplace presentations and client meetings.
- Individuals experiencing mild speech difficulties (mild stuttering, pronunciation problems, speech anxiety) seeking a judgment-free practice environment; this system is not intended as a medical diagnosis or a substitute for speech therapy.
- Educators and communication coaches who may use SpeechMate as an additional tool to monitor student progress across multiple sessions.

1.4.2 Functional Scope

- AI conversation training, mock interview simulations, presentation training and pronunciation training.

- Speech analysis: fluency, pronunciation, filler word detection and basic stuttering pattern detection.
- Presentation behavior analysis: eye contact, facial expressions, posture and confidence estimation.
- AI feedback: conversation guidance and personalized recommendations, delivered after each session.
- Progress monitoring: training history, performance dashboard and PDF report generation.

1.4.3 Project Limitations

- Supports English and Malay, including English–Malay code conversion only.
- Optimized for desktop and laptop browsers; no dedicated mobile app.
- Focused on mild speech difficulties; the system is explicitly not a medical diagnostic tool.
- Stuttering detection is limited to speech pattern analysis (repetition, long pauses) rather than clinical classification.
- Requires a working microphone and webcam; conversational AI component requires an internet connection to access the Gemini API.

1.5 Relevance of Project

Academically, SpeechMate contributes an integrated multimodal framework for communication training that combines speech recognition, conversational AI, and computer vision in a single assessment pipeline, along with a combined view-influency metric and a cognitive load-informed delayed feedback design that extends the single-modality speech or vision tools described in Chapter 2.

Practically, this project is directly driven by a documented real-world employability problem in the Malaysian graduate labor market. The working SpeechMate prototype offers students and job seekers a low-cost, personalized, and repeatable way to practice interviews

and presentations; provides lecturers and communication coaches with an additional, objective, data-driven tool to monitor student progress; and demonstrates concrete applications of accessible AI technologies to communication skill gaps that are not adequately addressed by current commercial MALL tools and presentation coaching.

1.6 Project Feasibility

The project is planned in two phases, FYP I and FYP II, and is expected to be completed by December 2026. FYP I focuses on domain understanding, literature review, data set collection, system architecture design, and initial AI model training. FYP II will focus on front-end and back-end development, system integration and validation, and final prototype deployment.

The project is technically feasible given the maturity and accessibility of the selected tools. Faster-Whisper, MediaPipe, and OpenCV are open source and optimized for CPU/GPU-efficient local inference, which supports the project’s goal of local deployment without requiring paid cloud infrastructure for speech and vision channels; only the conversation guidance component relies on external APIs (Gemini Flash). The project is scoped to be completed by a single developer using a standard laptop, open source frameworks, and low-cost or free-tier API access, which ensures resource requirements are within the reach of the final year of the project. Key risks – limited Malaysian accent speech data, reliance on external conversation APIs and technical complexity of real-time multimodal fusion – are mitigated through modular development based on sprints and prioritization of core features over secondary improvements, as detailed in the Gantt chart in Section 3.4.

CHAPTER 2

LITERATURE REVIEW

This chapter presents the theoretical foundations that informed the design of SpeechMate, followed by four closely related technical works that directly shape the architecture and evaluation approach of the proposed system.

2.1 Theoretical Foundations

2.1.1 Cognitive Load Theory

Sweller (1988) proposed that human working memory has a very limited capacity; when processing demands (intrinsic load) and environmental distractions (extrinsic load) exceed this threshold, cognitive load occurs and verbal performance deteriorates. This theory directly justifies two core design decisions of SpeechMate: the Pre-Presentation Script Assistant, which reduces intrinsic load before the session begins, and the Delayed Post-Session Feedback design, which eliminates the extrinsic load created by real-time on-screen cues during speech.

2.1.2 Automatic Speech Recognition Foundations

Dennis (2024) describes Automatic Speech Recognition (ASR) as the fundamental technology that converts spoken language into text, with modern systems using deep learning models trained on large-scale speech datasets to recognize words, phonemes and sentence structure with high accuracy. This supports the SpeechMate Speech Analysis module, which relies on accurate transcriptions as input for downstream fluency, filler word and disfluency analysis.

2.1.3 Dual-Coding Theory

Mayer (2001) proposed that the human brain processes information through two separate channels – the visual channel and the auditory/verbal channel – and that learning and

performance are optimized when these channels work in harmony rather than competing for the same sensory resources. This theory is central to the justification for eliminating on-screen text prompts during active speech in SpeechMate, because forcing the user to read a script (the visual channel) and speak fluently (the verbal channel) simultaneously creates channel competition that the system’s script-before-you-speak and delayed feedback design aims to avoid.

2.1.4 The Feedback Loops Framework

Carless (2006) and Butler and Winne (1995) emphasize that effective feedback must be timely, structured, and non-intrusive to support learning rather than trigger defensive or anxious responses. This underlies SpeechMate’s AI Feedback and Recommendations module, which aims to combine speech and visual analysis into a single performance score delivered after a session, along with structured, actionable improvement suggestions rather than undifferentiated raw metrics.

2.2 Project-Related Technical Works

2.2.1 AI-Based Speaking Assistant

Qin, Zhu, Yamashita, Yang, Suga, and Lee (2025) propose an LLM-based speech assistant that provides real-time text suggestions as a speech reference for non-native speakers in multilingual communication. This approach helps to structure the logical flow and depth of thought in real time, but its reliance on keyword prompts typed on the screen during active speech increases cognitive load and carries the risk of language hallucinations. For SpeechMate, this encourages the use of passive and multimodal tracking during training sessions, where the system observes and records the user’s speech and behavior rather than presenting direct text prompts that can distract the speaker.

2.2.2 FluentNet: End-to-End Detection of Stuttered Speech

Kourkounakis, Hajavi, and Etemad (2021) combine a Pinch-and-Excite residual convolutional network, a bidirectional LSTM, and an attention mechanism to detect end-to-end stuttering speech disfluency on the UCLASS corpus, achieving high accuracy by using the attention mechanism to capture key temporal regions of the speech signal. The main limitation of this model is its heavy reliance on data such as LibriStutter, which fails to fully capture real psychological speech blocks. For the SpeechMate Speech Analysis module, this drives the use of an attention layer architecture to help isolate where Malaysian accent filler words or stuttering spikes occur over time, while assuming that any locally trained model requires additional real speech data, rather than purely synthetic ones.

2.2.3 The Impact of Forced-Alignment Errors on Automatic Pronunciation Evaluation

Mathad, Mahr, Scherer, Chapman, Hustad, Liss, and Berisha (2021) examined how forced alignment errors affect Goodness of Pronunciation (GOP) scoring, applying the analysis to speech from typically developing children and children with cerebral palsy. Using a bivariate linear model, they found that while automatic forced alignment boundary errors on the order of 30–60 milliseconds occur, alignment errors overall are relatively weak predictors of the resulting pronunciation scores, with significant effects observed particularly for plosive and fricative phonemes. This supports the use of GOP-style forced alignment scoring as a relatively robust basis for the pronunciation feedback component of SpeechMate, while suggesting that scoring thresholds should be deliberately relaxed rather than strict, to ensure that the training experience encourages users, especially those with speech anxiety, rather than penalizing them for small alignment-related scoring noise.

2.2.4 A Real-Time Wearable Assistant for Improving Eye Contact in Presentations

Du, Zhang, and Lan (2026) present SpeakAssis, a real-time wearable system using a head-mounted eye tracker that analyzes the speaker’s gaze distribution across audience and non-audience areas and delivers audio prompts to correct ineffective eye contact patterns; in a user study involving eight speakers and twenty-four audience members, the system increased eye contact duration by an average of 62.5%. While effective, this approach relies

on dedicated wearable eye tracking hardware and delivers real-time voice prompts during active speech, which risks the same cognitive load and speech interruption concerns raised in Section 2.1.1. For SpeechMate, this prompts a reimplementaion of the same basic three-zone gaze distribution concept using only a standard webcam via MediaPipe, and silently records the resulting gaze data for post-session Gaze Tunneling composite metrics rather than interrupting the user in real time.

2.3 Research Gaps and Positioning

Across the literature reviewed, there are three consistent gaps. First, existing speech assistant tools intervene during active speech in a way that increases cognitive load rather than reducing it. Second, stuttering and pronunciation detection models are typically trained and validated on Western-accented or synthetic corpora rather than Malaysian-accented English or code-switched Malay-English speech. Third, performance behavior tools that achieve robust eye contact outcomes rely on specialized wearable hardware rather than standard webcams. SpeechMate is positioned to address all three gaps simultaneously through its delayed feedback design, the use of planned local and additional training data (Chapter 4), and the multimodal fusion channel of a webcam only.

2.4 Summary of Literature Review

Table 2.1 consolidates the eight works reviewed in this chapter, summarising the core contribution of each and how it is applied within SpeechMate.

Table 2.1: Summary of Literature Reviewed

Author (Year)	Core Contribution	Application to SpeechMate
Sweller (1988)	Cognitive Load Theory: working memory collapses under excess intrinsic/extraneous load	Motivates the Pre-Presentation Scripting Assistant and Deferred Feedback design
Dennis (2024)	Foundational overview of Automatic Speech Recognition (ASR)	Underpins the Speech Analysis module's reliance on accurate transcription
Mayer (2001)	Dual-Coding Theory: visual and verbal channels compete for cognitive resources	Justifies eliminating on-screen scripts during active speech

Carless (2006); Butler & Winne (1995)	Feedback must be timed and structured to support learning, not overwhelm it	Grounds the AI Feedback and Recommendation module's single post-session score
Qin et al. (2025)	LLM-driven real-time speaking suggestions for non-native speakers	Motivates passive, silent multimodal tracking instead of live text prompts
Kourkounakis et al. (2021)	FluentNet: CNN + BiLSTM + attention for end-to-end stutter detection	Basis for SpeechMate's proposed stuttering/disfluency detection architecture
Mathad et al. (2021)	Forced-alignment error is a weak predictor of GOP pronunciation score	Justifies relaxed GOP acceptance thresholds for anxious users
Du, Zhang & Lan (2026)	Wearable tri-zone gaze tracking improves presentation eye contact	Re-implemented webcam-only for the Gaze Tunneling fusion metric

CHAPTER 3

METHODOLOGY

3.1 Research Methodology

SpeechMate was developed using an Agile methodology structured around six functional accelerators, which allowed each core module to be designed, implemented, and validated in stages rather than as a single monolithic build. This approach was chosen because it allowed AI-intensive modules (speech analytics, computer vision) to be iterated independently of more conventional web application modules (user management, dashboards), and allowed scope to be adjusted if a particular model or data set proved more difficult to obtain than initially planned.

1. Sprint 1 – User Management Module: authentication, profile management, progress tracking and personal settings.
2. Sprint 2 – Conversational AI Module: intelligent dialogue generation, communication scenario handling and speech training flow.
3. Sprint 3 – Speech Analysis Module: speech recognition, fluency assessment, stuttering detection and filler word analysis.
4. Sprint 4 – User Behavior Module: computer vision-based eye contact analysis, facial expression recognition and posture assessment.
5. Sprint 5 – Analytics Module: progress monitoring, performance visualization and recommendation generation.
6. Sprint 6 – Testing and Validation: functional testing, debugging and user testing to verify the platform is usable and reliable.

At the time of this writing, Sprints 1 through 3 are in early design and prototyping, consistent with the project being placed between the System Architecture and AI Model Training phases in the overall schedule (Section 3.4).

3.2 System Design

This section presents the overall system architecture and use case diagrams that together define how users and external services interact with SpeechMate..

3.2.1 System Architecture

SpeechMate follows a layered architecture, as illustrated in Figure 3.1. Learners interact with the system through a System Interface, which sends requests to the AI Guidance Engine, which consists of a Speech and Fluency Model and a Gaze-Behavior Fusion Model; the Engine returns personalized feedback and suggestions through the same interface. Below the Engine, the Feedback and Reporting System combines the Delayed Feedback Generator with an Analytics Dashboard, which is accessible to Educators and Administrators through a second System Interface. On the right, the Data Component stores Speech and Video Recordings, Session Metadata, and a User Progress Database, each connected by a discrete data flow connector to its corresponding processing component: recordings feed the Speech and Fluency Model, session metadata feed the Gaze-Behavior Fusion Model, and the Progress Database feed the Feedback and Reporting System.

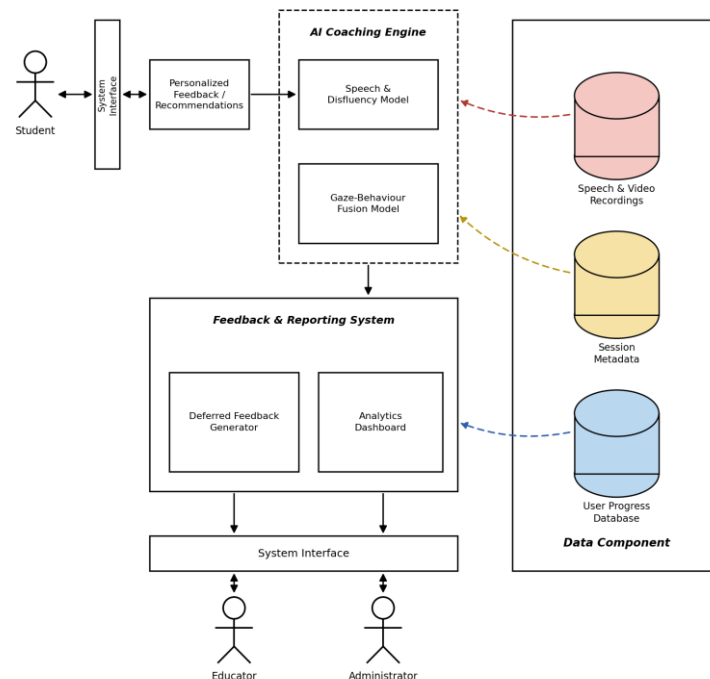


Figure 3.1: SpeechMate System Architecture Diagram

3.2.2 Use Case Diagram

Figure 3.2 shows the key actors and use cases for SpeechMate. Students receive AI-Based Registration and Login, AI Conversation Practice, and Delayed AI Feedback and own the resulting PDF Report and User Account. In this system, AI Conversation Practice triggers Speech and Behavior Analysis, which in turn generates Delayed AI Feedback, which reflects the post-session-only feedback design of the project. Educators monitor student progress, based on the same Performance Dashboard that Students receive and provide scenarios and training content used during AI Conversation Practice. Administrators manage a centralized user database and system configuration, which in turn controls and owns the use cases of User Accounts shared with Students.

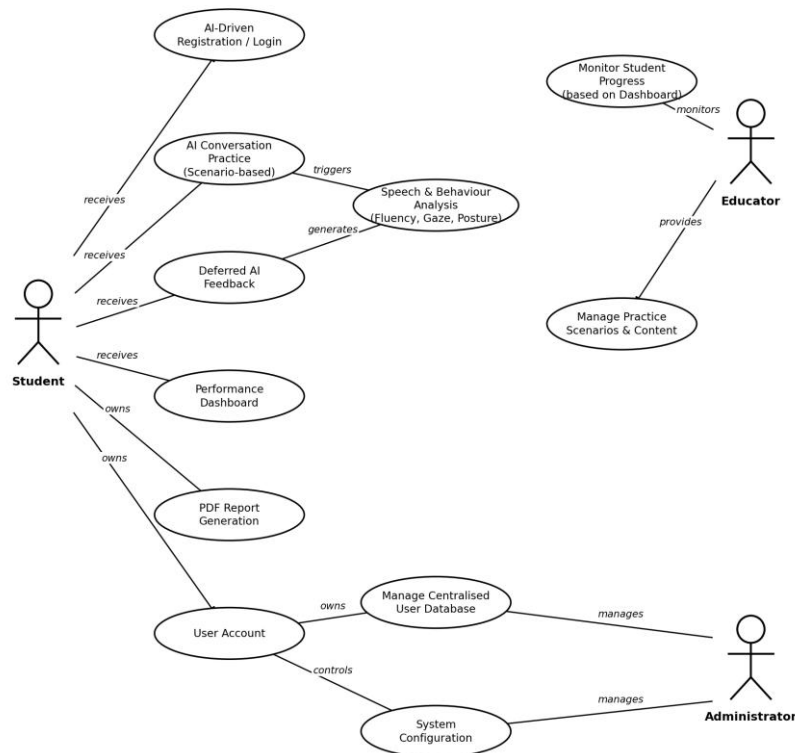


Figure 3.2: SpeechMate Use Case Diagram

3.3 Tools

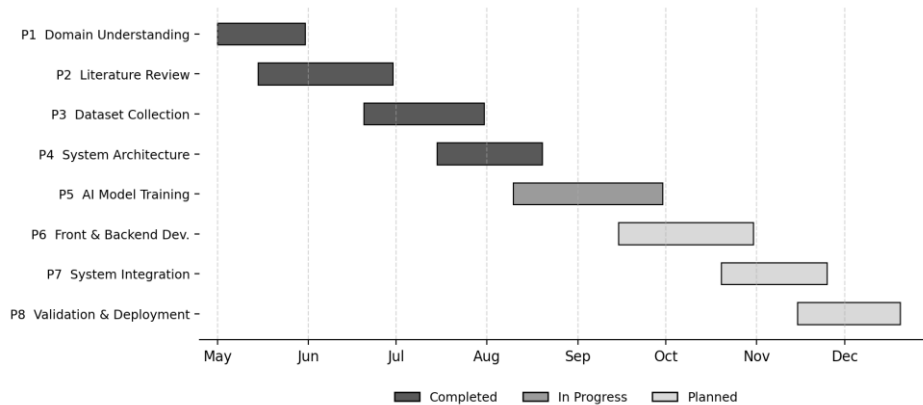
SpeechMate is built on a free or open source framework with local deployment in mind, minimizing infrastructure costs while ensuring that each layer is interchangeable during development. Table 3.1 summarizes the technology stack by layer.

Table 3.1: SpeechMate technology stack by layer

Layer	Tools / Technologies
AI / LLM	Faster-Whisper (ASR), Gemini Flash (LLM), PyTorch, MediaPipe, OpenCV, GOP + Wav2Vec2, CNN + BiLSTM with attention
Data & Knowledge	PostgreSQL, MinIO (object storage), Parquet
Backend	FastAPI, Python, SQLAlchemy, JWT authentication, Celery (asynchronous task queue)
Frontend	Next.js, TailwindCSS, Zustand (state management), Recharts (charting), React Webcam, Web Audio API
Live Data & API	Gemini API, MediaPipe API, WebRTC
Infrastructure	Docker Compose, Nginx, GitHub (version control)

3.4 Gantt Chart

The project is divided into eight phases spanning May to December 2026, structured across FYP I and FYP II. This interim report is submitted at the end of Phase 4 (System Architecture), with Phase 5 (AI Model Training) currently underway, as illustrated in Figure 3.3.

**Figure 3.3: Project Gantt Chart (May – December 2026)**

3.5 Expected Outcome

The completed system is expected to provide a single integrated dashboard built from six functional modules, which are summarized in Table 3.2.

Table 3.2: SpeechMate's six core modules and expected functionality

Module	Name	Function Summary
M1	User Management	User registration/login, profile and goal storage, session tracking, personal learning records
M2	AI Communication Practice	Microphone-based interaction with the conversational AI across casual, interview, and presentation scenarios
M3	Speech Analysis	Converts speech to text; measures speaking rate; detects filler words, pauses, and stuttering indicators
M4	Presentation Behaviour Analysis	Webcam-based eye contact, facial expression, and posture analysis to estimate confidence
M5	AI Feedback & Recommendation	Combines speech and visual analysis into a single performance score with improvement suggestions
M6	Progress Monitoring & Reporting	Stores session results, tracks improvement over time, generates downloadable PDF reports

3.6 Process Flowchart

Figure 3.4 illustrates the overall process flow of the system from the perspective of each actor. After the process begins, the system determines the current user role. If the user is a Student, the flow will proceed to scenario selection, AI conversation training, speech and video recording, and, optionally, requesting assistance from an AI chatbot. If the user is an Educator instead, the flow will proceed to the preparation of assessment criteria and training content. If the user is an Administrator, the flow will proceed to the management of student and educator databases and system configuration. All three role-based paths meet at the Dashboard Analysis stage, where speech patterns, behavior, and usage are analyzed together. This feeds into the Suggestions and Delayed Feedback for Students stage, and the Content Review / Enhance Lesson Plan stage that allows Educators and Administrators to update training materials based on the aggregate results, before the process ends.

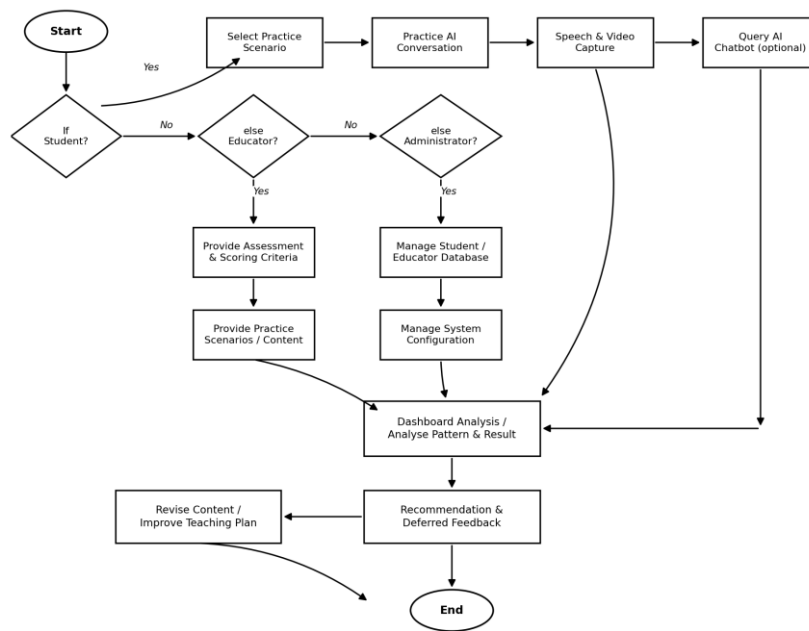


Figure 3.4: SpeechMate System Process Flowchart

CHAPTER 4

FINDINGS AND DISCUSSION

***Note on scope:** as this project is moving from system architecture (Phase 4) to AI model training (Phase 5), no models have been trained or evaluated on real data at the time of this writing. Therefore, this chapter presents the planned dataset, preprocessing pipeline, and training and evaluation configurations, which are derived directly from the literature reviewed in Chapter 2 and from the technical evaluation framework defined at the proposal stage. The actual training results, evaluation metrics, and quantitative findings will be reported in the final project report once the activities of Phases 5 through Phase 8 are complete.*

4.1 Data Collection

Three complementary data sources are planned to address the lack of Malaysian accent and code-switching data identified in Section 2.3.

First, public benchmark corpora will be used, including UCLASS and SEP-28K for stuttered and non-fluent speech (used by Kourkounakis et al., 2021), L2-ARCTIC and LibriSpeech for modeling non-native pronunciation (used in the framework of Nijat, Wei, Li, Dawut & Hamdulla, 2026), and general-purpose ASR benchmarks for basic Whisper evaluation. Second, locally recorded Malaysian English and English-Malay code-switching speech samples will be collected from consenting volunteer participants, covering casual conversations, mock interviews, and presentation style prompts, to calibrate the system to the target accent population rather than a purely native norm basis. Third, additional synthetic data can be generated following the Stutter-TTS approach of Zhang et al. (2022), where controlled synthetic disfluency is injected into standard Malaysian accented reading transcripts using a local text-to-speech engine, expanding the amount of labeled stuttering data without the need for a large manually annotated diagnostic corpus.

4.2 Data Preparation and Preprocessing

Audio preprocessing is designed to include noise reduction, such as spectral rejection or gating, so that the system works reliably through standard laptop microphones rather than studio-grade equipment. Transcripts generated by Faster-Whisper will be time-stamped on the corresponding video stream so that speech events, including fillers, stutters, and pauses, can be correlated with simultaneous gaze and posture events during the fusion stage. Video preprocessing will use MediaPipe facial and pose landmark extraction to derive normalized gaze direction vectors and shoulder alignment angles per frame, rather than storing raw video, to reduce storage overhead. The labeling of the locally collected sample set is designed to use a lightweight manual annotation pass, marking filler words, repetitions, and prolonged pauses, cross-checked with automated Whisper transcripts, consistent with the evaluation approach summarized in Table 4.2.

4.3 Datasets and Training Configuration

Table 4.1 summarizes the components of the proposed model, the training and reference data planned for each, and the literature sources that motivated the choice of configuration.

Table 4.1: Proposed model components and training configuration

Component	Proposed Configuration	Literature Basis
Speech-to-text	Faster-Whisper (base/small model) run locally; fine-tuning to be revisited if Malaysian-accent word error rate proves high	Dennis (2024)
Stuttering / disfluency detection	CNN + BiLSTM with attention mechanism, trained on UCLASS/SEP-28K and locally augmented data	Kourkounakis et al. (2021)
Pronunciation scoring	GOP forced-alignment scoring with deliberately relaxed acceptance thresholds	Mathad et al. (2021)
Eye contact / gaze tracking	MediaPipe face-mesh landmark extraction mapped to a tri-zone (left/right/slide) gaze-distribution metric, webcam-only	Du, Zhang & Lan (2026)
Posture / facial expression	MediaPipe Pose with a lightweight facial-expression classifier (neutral/happy/nervous/confident)	Project scope
Conversational coaching	Gemini Flash via API, prompted per scenario (casual/interview/presentation)	Project scope

4.4 Model Evaluation

Model evaluation will follow the nine-component technical evaluation framework defined at the proposal stage, summarized in Table 4.2. Each AI component will be benchmarked against either a recognized metric, such as Word Error Rate, or manually annotated ground truths collected from local sample sets.

Table 4.2: Planned technical evaluation framework

Component	Evaluation Method	Metric
Speech Recognition	Compare AI-generated transcript with actual spoken text	Word Error Rate (WER)
Fluency Analysis	Compare AI analysis with manual observation	Speaking Rate (WPM), pause-detection accuracy
Pronunciation Assessment	Compare pronunciation scores with human evaluation	Pronunciation accuracy score
Filler Word Detection	Compare detected fillers with manually counted fillers	Detection accuracy
Basic Stuttering Detection	Compare detected repetitions/pauses with manual annotation	Precision, Recall, F1-score
Eye Contact Detection	Observe gaze direction compared with actual video	Detection accuracy
Posture Detection	Compare posture classification with manual observation	Classification accuracy
Facial Expression Analysis	Compare detected expressions with observed expressions	Emotion classification accuracy
Overall Communication Score	Compare AI-generated score with supervisor/expert rating	Correlation analysis

In addition to component-level testing, the completed system will also undergo functional testing (module-by-module verification of requirements), user acceptance testing (task completion and satisfaction questionnaires), and performance and reliability testing (response time and resource usage under typical laptop hardware).

4.5 Discussion of Model Results

Several of the design decisions in Table 4.1 are direct responses to the limitations identified in Chapter 2 rather than default choices. The decision to relax the GOP acceptance threshold, for example, is a direct follow-up to Mathad et al.'s (2021) finding

that alignment errors are relatively weak predictors of pronunciation scores, suggesting that a strict binary pass/fail threshold would penalize users for measurement noise rather than genuine mispronunciation, a pattern that this project was designed to avoid given its focus on users with speech anxiety. Similarly, reimplementing Du, Zhang, and Lan's (2026) three-zone gaze metric using only a standard webcam, rather than the wearable eye tracker used in the original study, trades some tracking accuracy for zero additional hardware cost and full alignment with the scope of the project's browser-based local use.

The main anticipated risks at this stage are the lack of data for Malaysian accented English and code-switching English, which none of the public corpora reviewed in Chapter 2 directly address. The three-pronged data collection strategy in Section 4.1 aims to mitigate this risk, but its actual adequacy can only be confirmed once Phase 5 training has commenced and initial Word Error Rate and F1 figures are available. These will be reported quantitatively in the final project report.

4.6 Interface Design: Wireframe and Storyboard

In preparation for the full front-end implementation in Phase 6, a low-fidelity wireframe and user journey storyboard were generated to validate the interaction model described in Chapters 2 and 3, particularly the requirement that no scores, alerts, or timer cues appear on the screen while the user is actively speaking.

Figure 4.1 shows two wireframe screens. The left panel is the Training Session screen: a live webcam preview without overlays, an active recording indicator, the selected scenario, and a silent elapsed time counter that does not distract the user; the AI interviewer prompts are shown as normal conversational text rather than coaching scripts. The right panel is the Performance Dashboard that is shown only after the session ends: four summary scorecards (fluency, pronunciation, eye contact, confidence), a radar chart comparing the current session to previous sessions, a list of session history, AI coach feedback text, and a button to download a PDF report. Intentionally, no equivalent scoring panel exists on the Training Session screen, which is a wireframe-level expression of the Delayed Feedback design (Section 2.1.1).

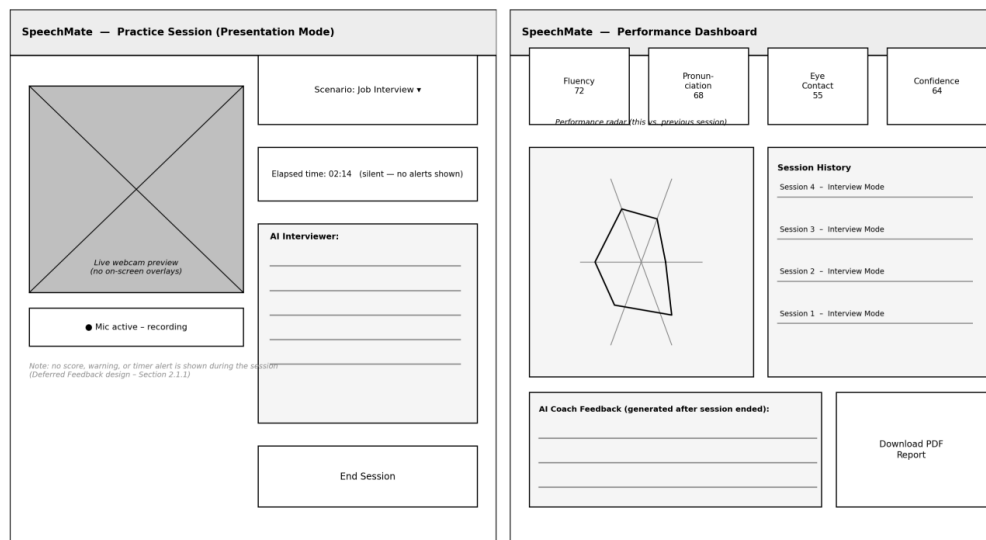


Figure 4.1: SpeechMate Wireframe Mockups (Practice Session & Dashboard)

Figure 4.2 traces the same interactions as the six-step storyboard: the user logs in and selects a training scenario; the session begins with the webcam and microphone active; the user speaks without interruption while the system silently records; the session ends and processing runs in the background; feedback and pending scores are only revealed at this point; and the user reviews dashboard trends and downloads a PDF report. This storyboard was used to check, before any implementation efforts, that each screen transition in the six-module design (Chapter 3) is mapped to a single, concrete user action, and that no steps require the system to interrupt the ongoing session.

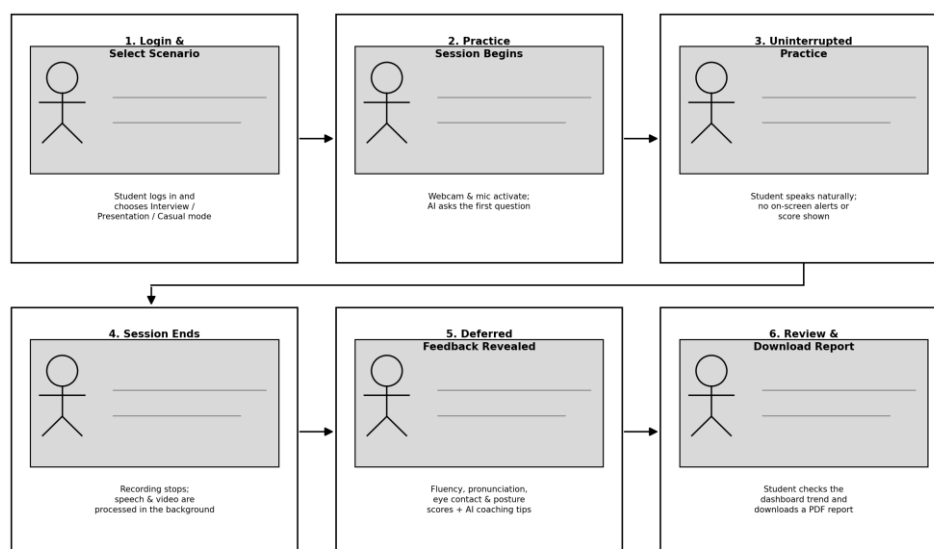


Figure 4.2: SpeechMate User Journey Storyboard

A set of full-resolution wireframe screens is provided in Appendix A. These wireframes will be refined into high-fidelity Next.js/TailwindCSS components during Phase 6 (Front-End and Back-End Development); any deviations from this interaction model at implementation time will be documented and justified in the final project report.

CHAPTER 5

CONCLUSION

5.1 Summary of Project Progress

This project aims to address a specific limitation in AI-assisted communication training: conventional tools assess verbal fluency and nonverbal delivery behaviors, such as eye contact, as separate and disconnected cues, and typically interrupt speakers with real-time on-screen warnings while they are still speaking rather than relating these cues afterwards. Progress to date has established the necessary foundation for testing SpeechMate’s alternative approach, encompassing problem definition, literature review, system and interface design, and assessment planning.

The literature review in Chapter 2 confirms that this gap is recognized but not fully resolved. Existing LLM-driven speech assistants and forced alignment pronunciation scorers each improve the structuring of phone-level fluency and accuracy, but remain single-modal in their decision logic, while wearable gaze-tracking tools that perform correct eye contact rely on specialized hardware and deliver corrective prompts during active speech rather than afterwards. This lays out the project’s contribution: a multimodal webcam-only fusion channel that mathematically relates gaze coordinates to vocal stuttering timestamps on a unified timeline (Gaze Tunneling metric), coupled with a delayed post-session feedback design based on Cognitive Load Theory rather than a real-time intervention model.

Chapter 3 translates this direction into a concrete methodology, defining a full pipeline from webcam and microphone capture through Faster-Whisper transcription, MediaPipe-based gaze and posture extraction, multimodal fusion, Gemini-based delayed feedback generation, and dashboard and PDF reporting, supported by a six-layer system architecture and use case diagrams that encompass the roles of Learner, Educator, and Administrator. The chapter also outlines the evaluation approach, using Word Error Rate, precision, recall, F1 score, and classification accuracy or correlation across nine technical components to evaluate the system’s performance once implemented.

Chapter 4 identifies the planned data sources – public corpora (UCLASS, SEP-28K, L2-ARCTIC) augmented with locally recorded Malaysian accent samples and Gagap-TTS-style synthetic augmentation – specifically chosen because none of the public corpora reviewed in Chapter 2 capture Malaysian accented Malay-English disfluency patterns or code-switching. The stuttering detection and GOP pronunciation scoring configurations to be applied to this data were defined with deliberately relaxed acceptance thresholds, informed by Mathad et al.’s (2021) finding that forced alignment errors are poor predictors of pronunciation scores, so as not to penalize anxious speakers due to measurement noise. A low-fidelity wireframe and six-step user journey storyboard (Section 4.6) were also generated to verify that the delayed feedback interaction model remained at the interface level before implementation began. The main challenges anticipated at this stage – the lack of data for Malaysian accented speech and code-switching – and corresponding mitigation strategies are discussed in Section 4.5. At this stage, the project has completed problem formulation, literature review, system architecture and use case design, and interface wireframing. Data set collection and preprocessing, AI model training and fine-tuning, front-end and back-end integration, and full system evaluation remain outstanding and form the core of the next phase of the project..

5.2 Future Work

The remaining project phases will focus on the following:

1. Phase 5 Solution – collect and preprocess a set of local Malaysian accented speech and video samples, train and refine the stuttering detection and gaze detection components, and record the initial Word Error Rate, precision, and recall figures against the framework in Table 4.2.
2. Phase 6 – develop user management, validation, and reporting modules, and integrate the Next.js front-end with the FastAPI back-end.
3. Phase 7 – integrate the speech and computer vision processing pipeline with the analytics dashboard, and conduct unit and user acceptance testing.
4. Phase 8 – conduct the full technical, functional, and user acceptance evaluation described in Chapter 4, analyze the results against the proposed metrics, and deploy the validated final prototype.

The final project report will present the completed system, quantitative model evaluation results, user acceptance test findings, and a critical discussion of how the delivered system performance compares to the design objectives and goals derived from the literature outlined in this report.

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APPENDICES

Appendix A: SpeechMate Wireframe Mockups (Full Set)

The wireframes below are the same Practice Session and Performance Dashboard screens introduced in Figure 4.1 (Section 4.6), reproduced here at a larger size for detailed reference.

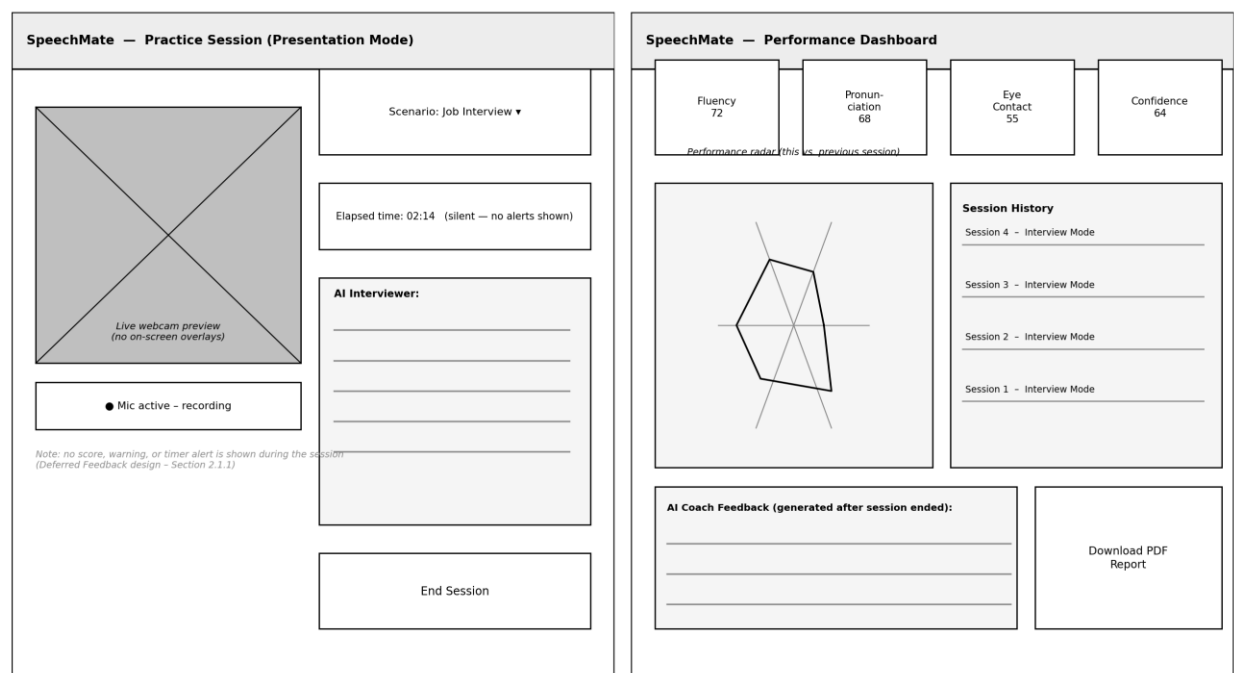


Figure A.1: SpeechMate Wireframe Mockups – Practice Session (left) and Performance Dashboard (right)

Appendix B: Draft User Acceptance Testing (UAT) Questionnaire

The following draft questionnaire is designed for the User Acceptance Testing activities in Phase 7 (System Integration) and Phase 8 (Validation and Implementation). Each item will be rated on a 5-point Likert scale (1 = Strongly Disagree, 5 = Strongly Agree), followed by two open-ended questions. This questionnaire will be refined once a working prototype is available for testing.

1. The registration and login process is easy to complete.
2. I understand how to select a training scenario (casual / interview / presentation).
3. During the training session, I do not feel distracted by on-screen warnings or scores.
4. The AI conversation feels natural and relevant to the selected scenario.
5. I find the delayed feedback (shown after the session) easy to understand.
6. The performance dashboard clearly shows my fluency, pronunciation, eye contact, and confidence.
7. I will trust the AI-generated feedback to guide my training.
8. I feel less anxious when practicing with SpeechMate than when practicing in front of a real audience.
9. I would use SpeechMate again to prepare for interviews or presentations.
10. The downloaded PDF report is clear and useful for reviewing my performance later.

Open question 1: What did you find most useful about practicing with SpeechMate?

Open question 2: What would you change about the training session or feedback experience?